

PRACTICAL NO 9 MODULE 2

AIM: Chi-Square Test of Independence

OUTPUT:-

The screenshot displays the RStudio interface with the following components:

- Console:** Shows the R script execution. The script reads a CSV file, checks column names, creates a contingency table for Gender and Disease_Category, and performs a Chi-Square test. The output shows the contingency table and the test results: $\chi^2 = 9.2779$, $df = 10$, $p\text{-value} = 0.5059$.
- Environment:** Lists the objects in the global environment, including 'data' (12000 obs. of 11 variables) and 'chi_result' (List of 9).
- Files:** Shows the file explorer with various datasets and scripts, including 'hospital_patient_dataset.csv'.

```
R - R 4.5.2 - C:/Users/as993/DATA SET/ <
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source
Console Terminal Background Jobs
> data <- read.csv("hospital_patient_dataset.csv")
> # Check the columns to be sure
> colnames(data)
[1] "Patient_ID"      "Hospital_Name"    "City"      "Gender"
[5] "Age"            "Disease_Category" "Admission_Type" "Length_of_Stay_Days"
[9] "Treatment_Cost"  "Insurance_Covered" "Discharge_Status"
> # Use 'Disease_Category' instead of 'Diagnosis'
> contingency_table <- table(data$Gender, data$Disease_Category)
> print(contingency_table)
      Cardiology Diabetes General Neurology Orthopedics Respiratory
Female    935      969    1004      946      992      932
Male     1040     1030     979    1052     1005     1000
Other       17       23       20       19       15       22
> # Run Chi-Square test
> chi_result <- chisq.test(contingency_table)
> chi_result

Pearson's Chi-squared test

data:  contingency_table
X-squared = 9.2779, df = 10, p-value = 0.5059
> |
```